

This assessment is designed to evaluate your research skills, technical adaptation, and understanding of modern Embodied AI workflows. Our goal is not just to see a working code snippet, but to observe how you approach new libraries (like LeRobot), how you handle documentation, and your grasp of fundamental robotics concepts like Inverse Kinematics (IK).

We are looking for your ability to troubleshoot, select appropriate models, and integrate distinct systems (Vision + Control).

Setup & Environment Guidelines

To ensure a smooth workflow, please adhere to the following recommendations:

- **OS:** Linux (Ubuntu 20.04/22.04) or Windows with WSL2 is highly recommended.
- **Environment Management:** Strictly use **Conda** or **venv** to isolate your dependencies.
- **Hardware:** VLA models (Task 1) benefit significantly from GPU acceleration. If you have an NVIDIA GPU, ensure your CUDA/PyTorch setup is correct. If you are restricted to a CPU, inference will be slower; plan your testing accordingly.
- **Visualization:** You may need tools like **ffmpeg** to capture and render simulation outputs.

Task 1: Embodied AI Exploration with SmoIVLA & LeRobot

Objective: Understand the architecture of Visual-Language-Action (VLA) models and deploy a pretrained model in a simulation environment.

Part A: Research & Concept

Perform a brief investigation into the following topics. Short, simple but your **own words** and view!

1. **LeRobot:** What is it, and why is Hugging Face building a unified library for robotics? How does it differ from traditional control stacks?
2. **SmoIVLA:** Research the general architecture. How does a VLA model differ from a standard LLM or Vision model in a robotics context?

Part B: Simulation & Inference

Using the **lerobot** library, you are required to run a **SmoIVLA** model inference on a robot arm within the **MuJoCo** simulation.

Requirements:

1. **Model Selection:** Navigate the Hugging Face Hub to find a suitable SmoIVLA model.
 - *Hint:* Avoid using base model, as they usually contain raw weights not optimized for specific robotic tasks. Look for **fine-tuned** checkpoints.
 - *Critical:* Pay attention to the dataset used to train the model. You must identify

which robot arm the model was trained on.

2. **Simulation Setup:** Use the **MuJoCo Menagerie** or **similar open source robotic arm(which xml and URDF files can be found online)** to load the robot arm that corresponds to your selected model.
 - *Note:* Using a model trained on an Aloha system to control a Panda arm will likely result in incoherent motions. Ensure your model and simulation robot match.
3. **Execution:** Run the inference loop. The goal is to see the robot arm perform coherent actions based on the model's policy.

Task 2: Vision-Based Teleoperation (Hand to Robot)

Objective: Demonstrate your ability to map human motion to robotic control using Inverse Kinematics (IK).

Scenario: You need to simulate a "pouring" motion. The input is a video of a human hand lifting a bottle and pouring; the output is a robot arm replicating this trajectory in simulation.

Requirements

1. **Robot Arm:** Use the **Trossen VX300s** (Right Arm) from the MuJoCo Menagerie.
2. **Input Data (Vision):**
 - Use **Google MediaPipe Hand Landmarker** or the **MANO** model to extract 3D landmarks from a video source.
 - You may record your own video (egocentric/first-person view is preferred) or video which we provide.
3. **Inverse Kinematics (IK):**
 - Map the extracted hand/wrist pose to the robot's End-Effector.
 - Implement an IK solver to calculate the necessary joint angles for the VX300s to mimic the hand's motion.
 - We encourage innovative solutions here. Don't expect one-shot solutions, since this is a complex, ongoing challenge. **We expect you to approach the problem with out-of-the-box ideas and truly leverage your expertise in computer vision and perception!**

Goal: The robot should mimic the lifting and rotation (pouring) motion of the hand in the simulation window.

Task 3: AI Tool Usage & Content Communication

Objective

This section evaluates how effectively you use AI tools during your workflow and how clearly you can communicate a technical robotics output to a broader audience.

We are not looking for generic AI-generated writing. We want to understand how you think, how you use AI tools to accelerate your work, and how well you can turn a technical result into clear content and presentation material.

Part A: AI Tool Usage Reflection

Briefly explain how you used AI tools during this case.

Please include:

1. Which AI tools you used
Examples: ChatGPT, Claude, Gemini, Cursor, Copilot, Perplexity, etc.
2. What you used them for
Examples: research, debugging, documentation understanding, code generation, error fixing, writing, or summarization.

Part B: Social Media Thread

Create one X/LinkedIn thread in English explaining your case output.

Requirements:

- Minimum 4–6 paragraphs
- The thread should explain what you attempted, what worked, what did not work, and why the result matters.
- The writing should be clear, concise, and understandable for a technical but not necessarily robotics-expert audience.
- Avoid generic marketing language. Focus on the real process, learnings, and implications.

Part C: Short Presentation

Prepare a short presentation in English summarizing your work.

Requirements:

- 5–7 slides
- Format: PDF, Google Slides, or PowerPoint
- The presentation should be clear enough for both technical and non-technical reviewers.
- Focus on your approach, decisions, challenges, results, and learnings.

Deliverables

Please organize your submission into a single repository or a zipped folder containing:

1. **Task 1 Report:** A brief document (approx. 1 page) covering your research findings, the specific SmolVLA model you selected (and why), and an evaluation of the inference results (what worked, what didn't, and technical reasons why).
2. **Task 1 Visualization:** A rendered video or GIF of the SmolVLA model controlling the robot in MuJoCo.
3. **Task 2 Visualization:** A side-by-side video (or combined view) showing the reference hand video and the robot simulation performing the motion simultaneously.
4. **Task 3 Social Media Content:** An English X thread or LinkedIn post explaining your case output and why it matters.
5. **Task 3 Presentation:** A short 5–7 slide presentation in English summarizing your approach, results, challenges, and key learnings.

Resources & Documentation

LeRobot & SmolVLA

- [GitHub: Hugging Face LeRobot](#)
- [Documentation: LeRobot Index](#)
- [Guide: SmolVLA Usage](#)

Simulation & Robots

- [GitHub: Google DeepMind MuJoCo](#)
- [GitHub: MuJoCo Menagerie](#)
- [Specific Robot: Trossen VX300s](#)

Computer Vision & Kinematics

- [Tool: MediaPipe Hand Landmarker](#)
- [Model: MANO Hand Model](#)
- [Concept: Inverse Kinematics](#)

Good luck! We look forward to seeing your approach.